

Submission Cover Sheet

Course Details

Title: Blockchain Fundamentals

Code: DAT5002 S1_25

Term: Autumn 2025

Assessment: **WRIT1**

Student Details

Tom Rowan ST20285213

Project Tutor

Dr. Ali Shahaab

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Section 1 - Theoretical understanding of the functioning of Blockchain Technologies and its applications in different sectors.

Task 1.1

A DLT is a Distributed Ledger Technology. This means that rather than a single centralised ledger or database there are multiple copies across many systems.

Type of DLT and Data Structures

Blockchain

A blockchain is based on a linked list of blocks, proceeding linearly. Miners or Validators propose new blocks. In the event of divergence, the longest resulting chain is followed; others are destroyed. Mining is computationally expensive (Nakamoto 2008).

Advantages

- High confidence and security.
- Transparency.

Disadvantages

- Scalability is limited, with transaction rates under 50 TPS.
- High energy consumption in Proof of Work systems.

Directed Acyclic Graph (DAG)

A DAG is a graph where links between nodes can only proceed in one direction, loops are avoided. Nodes can refer to multiple parent and children blocks, allowing parallel chains. Every user who submits a transaction also helps validate others, eliminating dedicated miners. (Bashir 2023).

Advantages

- High scalability.
- Very high throughput (1000s TPS).
- Lower (zero) transaction fees.
- Works with low-power hardware.

Disadvantages

- Potential security issues.
- Potential loss of consistency.

Hashgraph

Structured as a DAG. Uses “Gossip about Gossip” protocol and virtual voting consensus, saving bandwidth (Baird 2016).

Advantages

- Very high throughput (up to 500,000 TPS).
- Low latency.
- High Security using Asynchronous Byzantine Fault Tolerance.
- Low energy consumption.

Disadvantages

- Technology is patented (eg: Hedera Hashgraph).
- Decentralisation is limited.

Holochain

Each node (agent) in a Holochain network maintains its own chain, allowing decentralised coordination. This uses a Distributed Hash Table (DHT), which stores key value pairs. (Holochain 2020).

Advantages

- Lightweight and scalable.
- Enhanced privacy and control over data.

Disadvantages

- Limited adoption so far.
- More complex to implement.

Network Architectures

The different network architectures used for DLTs provide different levels of performance, scalability and decentralisation. Many DLTs use a Peer-to-peer network where each node has the same status and role, like the BitTorrent file sharing network.

Public Blockchain networks like Bitcoin or Ethereum are open participation networks where anyone can join, read data, validate a transaction or take part in the consensus process. No central authority is in control.

Private DLT networks require permission to join. Only specified individuals or nodes can join the network or read data from the ledger. They are completely owned and operated by a single organisation who sets the rules and governs the network.

Consortium DLTs are also limited access. However, they are managed cooperatively by multiple organisations, sharing governance among them.

The following table contrasts the advantages and disadvantages of these technologies.

DLT Type	Advantages	Disadvantages
Permissionless Public	Open to participation by anyone. Resistant to single points of failure. All transactions are visible and traceable. Distributed governance, resisting censorship and control. It is difficult to ban people or interfere with transactions. (Nakamoto 2008)	All transactions are visible and traceable. Slow transaction throughput due to massive scale synchronisation requirements. Very high electricity consumption for PoW. Transactions incur high fees.
Permissioned Private	High transaction throughput and efficiency. Fully controlled governance of network. Strong privacy. (Bashir 2023)	Fault tolerance risks due to centralisation. Limited trust outside of the controlling organisation.

Permissioned Consortium	Reasonable transaction throughput and efficiency. Suitable for multiple organisational projects. (Bashir 2023)	Complex governance model. Requires trust among consortium members.
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Task 1.2

A summary of the top five consensus protocols (Xu 2023, Microsoft 2025, DeepSeek 2025, OpenAI 2025):

Consensus Protocol	Who Participates	How Agreement is Reached	Advantages / Disadvantages
Proof of Work	Anyone with computing power (Miners)	Solving computationally expensive hashing puzzle to propose a block. Dynamic difficulty to preserve chain growth time. In contention use longest chain rule. (Nakamoto, 2009)	A: No centralised control. D: Uses a lot of energy.
Proof of Stake	Validators based on staked currency.	Putting the most 'skin in the game' allows validator to propose a block. Stake can be lost for bad behaviour.	A: Energy efficient. D: May end up with unintended centralisation if stakes are concentrated.
Practical Byzantine Fault Tolerance	Trusted Nodes	Nodes communicate in rounds to agree on the order of transactions. Requires known and trusted validators.	A: Tolerates up to one-third of faulty or malicious nodes. (Castro 1999)
Delegated Proof of Stake (DPoS)	Delegated Validators	Coin holders vote for delegates, who then validate blocks and transactions. (Larimer 2014).	A; High throughput. D: Potential delegate collusion.
Proof of Authority	Trusted Nodes	A limited number of trusted nodes are pre-approved to validate transactions. Strong authentication and trust of nodes is vital.	A: High throughput. D: Undemocratic centralised control.

[194 words]

Task 1.3

Financial Services

DLT technologies allow for faster, cheaper, and more secure international payments by removing intermediaries such as banks.

By eliminating intermediaries, blockchain based systems make it harder for governments or central authorities to control or freeze transactions, offering greater financial autonomy for individuals and businesses. This may simultaneously be a disadvantage as it allows sanctioned groups or individuals to participate. (Catalini 2016).

In a traditional system, intermediary banking systems (such as SWIFT) introduce both costs and delays. Blockchain allows payments to be processed significantly faster, and without percentage-based fees. Reduced transaction fees benefits both businesses and consumers. (Catalini 2016)

Transactions are verified using cryptography and are therefore reliable and immutable; they cannot be reversed or denied increasing trust in banking. (Nakamoto 2008).

Supply Chain Management

A Consortium-based blockchain can track products through the entire supply chain, from the original manufacturer or grower to the destination vendor. This ensures the authenticity of goods and makes counterfeiting challenging.

Increased transparency and reduced fraud can lead to increased consumer trust in a premium product. For example, in 2018, De Beers trialled a blockchain platform to track diamonds from the mine to the retail store (De Beers 2018). This allowed customers to track provenance, ensuring that no 'blood diamonds' are used.

Block chain increases information transparency and creates information sharing opportunities between trading partners. (Chang 2020).

Electronic Health Records

Many people use wearable technology to track health data, including both self-owned commercial devices, such as Fitbit or Apple Watch, and healthcare system provided devices which are increasingly used in clinical settings for remote monitoring. This data is highly valuable for monitoring ongoing conditions, enable personalised treatment plans, but is also vulnerable to security breaches and tampering (Tariq 2020).

In addition, most healthcare providers around the world now rely on computer-based medical records which may include not only traditional medical records but also

collections of data from wearables. This makes healthcare provision more efficient and effective but also increases the complexity and volume of sensitive patient information.

A consortium based blockchain system enables secure, immutable records of patient health data that can be shared across multiple healthcare providers, while maintaining transparency and data privacy.

The key benefits are a reduction in fraud, as the immutability of blockchain prevents unauthorised alterations or deletions of records; repudiation of such changes becomes impossible. It also allows improved access and secure sharing of data across otherwise incompatible computer systems. It also enables patients to control who can access their health information. (Tariq 2020).

[415 words]

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